FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 5 or 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

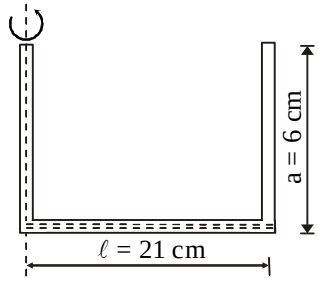
SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. In a large room a person receives direct sound waves from a source 120 m away from him. He also receives waves from the same source which reach him, being reflected from the 25 m high ceiling at point half way between them. For which wavelength will these two sound waves interfere constructively?
(A) 10m (B) 5m
(C) 8 m (D) 4 m

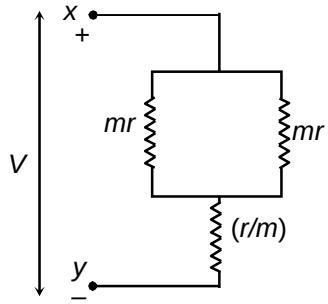
2. 4 moles of an ideal gas having γ = 1.67 are mixed with 2 moles of another ideal gas having γ = 1.4. Then which of the following is not the equivalent value of γ for the mixture.
(A) 1.54 (B) 2.34
(C) 1.23 (D) 1.87

3. Length of a horizontal arm of a U-tube is l = 21cm and ends of both of the vertical arms are open to surroundings of pressure 10500 Nm^{-2}. A liquid of density ρ =10^3 kg m^{-3} is poured into the tube such that liquid just fills horizontal part of the tube. Now, one of the open ends is sealed and the tube is then rotated about a vertical axis passing through the other vertical arm with angular velocity ω_0 = 10 radians/sec. If length of each vertical arm be a = 6 cm, calculate the length of air column in the sealed arm.



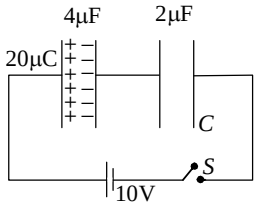
(A) 5cm (B) 4cm
(C) 7cm (D) 8cm

4. In the given circuit the value of *m* is varying. The correct statements about the circuit are
(A) The condition for maximum current flowing from *x* is *m* =2.
(B) The maximum current is $\frac{V}{\sqrt{2}r}$.
(C) The condition for maximum current flowing from *x* is $m = \sqrt{2}$.
(D) The maximum current is $\frac{3V}{2r}$.



Space for Rough work

5. A $4\mu F$ capacitor is given $20\mu C$ charge and is connected with an uncharged capacitor of capacitance $2\mu F$ as shown in figure. When switch *S* is closed.



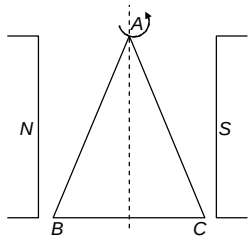
(A) Charged flown through the battery is $\frac{40}{3}\mu C$

(B) charge flown through the battery is $\frac{20}{3}\mu C$

(C) Work done by the battery is $\frac{200}{3}\mu J$

(D) work done by the battery is $\frac{100}{3}\mu J$

6. A coil in the shape of an equilateral triangle of side 0.02 m is suspended from vertex such that it is hanging in a vertical plane between the pole-pieces of a permanent magnet producing a horizontal magnetic field of 5 × 10–2T. Current in loop is 0.1 A. Then:



(A) Magnetic moment of the loop is $\sqrt{3}\times10^{-5}A-m^2$

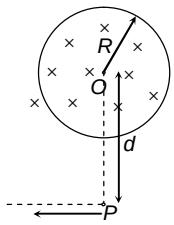
(B) Magnetic moment of the loop is $1\times10^{-6}A-m^2$

(C) Couple acting on the coil is $5\sqrt{3}\times10^{-7}N/m$

(D) Couple acting on the coil is $\sqrt{3}\times10^{-7}N-m$

Space for Rough work

7. In a cylindrical region of radius R, there exists a time varying magnetic field B such that $\frac{dB}{dt} = k(>0)$. A charged particle having charge q is placed at the point P at a distance d ($> R$) from its centre O. Now, the particle is moved in the direction perpendicular to OP (see figure) by an external agent upto infinity so that there is no gain in kinetic energy of the charged particle. Choose the correct statement/s.



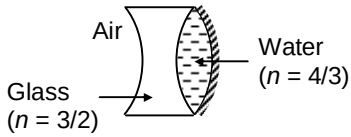
(A) Work done by external agent is $\frac{q\pi R^2}{4}k$ if $d = 2R$

(B) Work done by external agent is $\frac{q\pi R^2}{8}k$ if $d = 4R$

(C) Work done by external agent is $\frac{q\pi R^2}{4}k$ if $d = 4R$

(D) Work done by external agent is $\frac{q\pi R^2}{4}k$ if $d = 6R$

8. The radius of curvature of the left and right surface of the concave lens are 10 cm and 15 cm respectively. The radius of curvature of the mirror is 15 cm
(A) equivalent focal length of the combination is –18 cm
(B) equivalent focal length of the combination is +36 cm
(C) the system behaves like a concave mirror
(D) the system behaves like a convex mirror



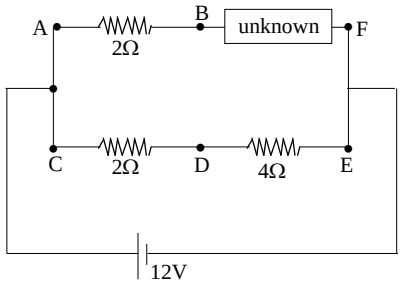
Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

Consider a circuit shown in figure.



List – I		List – II	
(I)	Current through A – B	(P)	Zero
(II)	Voltage across A – B	(Q)	12V
(III)	Current through B – F	(R)	4A
(IV)	Voltage across B – F	(S)	2A
		(T)	4V
		(U)	8V

9. If unknown box contains a resistor of 4Ω:

(A) $I \rightarrow S,\ II \rightarrow U,\ III \rightarrow P,\ IV \rightarrow P$

(B) $I \rightarrow S,\ II \rightarrow T,\ III \rightarrow S,\ IV \rightarrow U$

(C) $I \rightarrow R,\ II \rightarrow U,\ III \rightarrow R,\ IV \rightarrow T$

(D) $I \rightarrow S,\ II \rightarrow P,\ III \rightarrow S,\ IV \rightarrow U$

Space for Rough work

10. If the unknown box contains a $2\mu F$ capacitor then in steady state:

(A) $I \to P,\ II \to P,\ III \to P,\ IV \to Q$

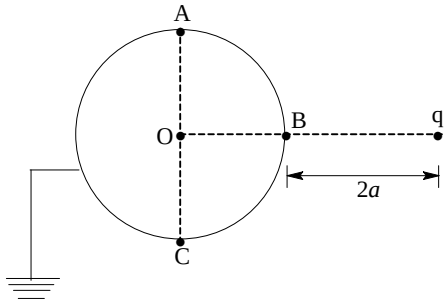
(B) $I \to P,\ II \to T,\ III \to P,\ IV \to U$

(C) $I \to R,\ II \to U,\ III \to R,\ IV \to T$

(D) $I \to S,\ II \to T,\ III \to S,\ IV \to U$

Answer the following by appropriately matching the list based on the information given in the paragraph.

Consider a conducting sphere as shown in figure.



Radius of sphere = a

List – I gives the four points.

List – II gives the values for potentials.

List – I		List – II	
(I)	A	(P)	Zero
(II)	B	(Q)	$\frac{-1}{4\pi\varepsilon_0}\frac{q}{3a}$
(III)	C	(R)	$\frac{-1}{4\pi\varepsilon_0}\frac{q}{a\sqrt{10}}$
(IV)	O	(S)	$-\frac{1}{4\pi\varepsilon_0}\frac{q}{2a}$
		(T)	$\frac{1}{4\pi\varepsilon_0}\frac{q}{3a}$
		(U)	None of these

11. If list – II gives the potential due to all changes then:

(A) $I \to P,\ II \to Q,\ III \to P,\ IV \to P$

(B) $I \to U,\ II \to U,\ III \to R,\ IV \to S$

(C) $I \to P,\ II \to P,\ III \to P,\ IV \to P$

(D) $I \to Q,\ II \to R,\ III \to U,\ IV \to P$

Space for Rough work

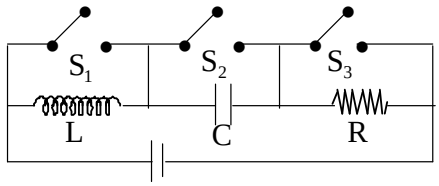
12. If list – II gives potential only due to surface charge (on the surface of the sphere), then:
(A) $I \to R,\ II \to S,\ III \to R,\ IV \to Q$
(B) $I \to Q,\ II \to P,\ III \to T,\ IV \to Q$
(C) $I \to P,\ II \to P,\ III \to R,\ IV \to Q$
(D) $I \to R,\ II \to Q,\ III \to T,\ IV \to U$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A positively charged particle starts at rest 25cm from a second positively charged particle which is held stationary throughout the experiment. The first particle is released and accelerates directly always from the second particle. When the first particle has moved 25cm, it has reached a velocity of $10\sqrt{2}$ m/s. The maximum velocity (in m/s) that the first particle will reach is $10 \times x$, find x?

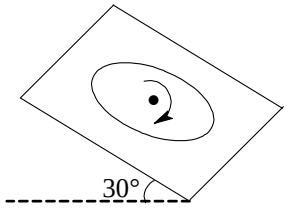
14. Consider the circuit shown in figure. With switch S_1 closed and the other two switches open, the circuit has a time constant 0.05 sec. With switch S_2 closed and the other two switches open, the circuit has a time constant 2 sec. With switch S_3 closed and the other two switches open, the circuit oscillates with a period T. Find T (in sec). (Take $\pi^2 = 10$)



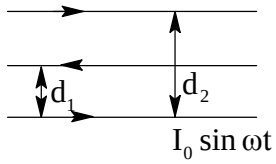
15. In a slow reaction, heat is being evolved at a rate about 10mW in a liquid. If the heat were being generated by the decay of ^{32}P, a radioactive isotope of phosphorus that has half-life of 14 days and emits only beta-particles with a mean energy of 700KeV. The number of ^{32}P atoms in the liquid is $A \times 77 \times 10^{15}$ find a:
[Take: ln2 = 0.7]

Space for Rough work

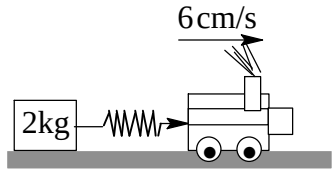
16. An old record player of 10 cm radius turns at 10 rad/s while mounted on a 30° incline as shown in the figure. A particle of mass m can be placed anywhere on the rotating record. If the least possible coefficient of friction μ that must exist for no slipping to occur is μ, find $2\sqrt{3}\mu$. [Take g = 10 m/s^2]



17. Two long straight wires of telephone circuit inside a house are coplanar with and run parallel to a third long straight conductor that carries a current $i = i_0 \sin \omega t$. The wires are at a distance of d_1 and d_2 from the conductor. The conductor induced a noise in the telephone circuit. The rms voltage per unit length of this induced emf is $x \times 5$, find x? Express your answer in μV. Take $d_2 = 2d_1, i_0 = \dfrac{2\sqrt{2}}{\ln 2} A\ \omega = 50\ rad/s$



18. Figure shows a spring attached to a 2.0 kg block. The other end of the spring is pulled by a motorized toy train that moves forward at constant speed 6.0 cm/s. The spring constant is 50 N/m and the coefficient of static friction between the block and the surface is 0.60. The spring is at its natural length at t=0s when the train starts to move. When does the block slip (express your answer in second)?



Space for Rough work

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following statement(s) is/are correct?
(A) Sodium bicarbonate is more soluble than sodium carbonate
(B) Sodium hydroxide is known as caustic soda
(C) Mg^{2+} ions are precipitated with the addition of NH_4OH in the presence of NH_4Cl
(D) CaO_2 is less stable than MgO_2

20. Froth floatation:
(A) is a physical method of separating mineral from the gangue
(B) is a method to concentrate the ore depending on the difference in wettability of gangue and the ore
(C) is used for the sulphide ores
(D) is a method in which impurities sink to the bottom.

21. Which is/are correct statements?
(A) $LiCl$ is more soluble in polar solvent (water) than $NaCl$
(B) K_{a_2} fumaric acid is more than K_{a_2} of Maleic acid
(C) The $O-O$ bond length in $O_2[AsF_4]$ is shorter than KO_2
(D) In $CF_2=C=CF_2$ molecule all the four fluorine atoms are in the same plane.

22. Which of the following statement(s) is/are correct?
(A) The coordination number of each type of ions in $CsCl$ crystal is 8.
(B) A metal that crystallizes in BCC structure has a coordination number of 12.
(C) A unit cell of an ionic crystal shares some of its ions with other unit cells.
(D) The length of the unit cell in NaCl is 552 pm. $\left(r_{Na^+}=95pm, r_{Cl^-}=181pm\right)$

Space for Rough work

23. Which of the following relations(s) is/are true for alkaline solution?

(A) $pH > \frac{pK_w}{2}$ (B) $pH > pOH$

(C) $pOH < \frac{pK_w}{2}$ (D) $pH < pOH$

24. Which of the following are characteristics of thermosetting polymers?
(A) Heavily branched cross linked polymers
(B) Linear slightly branched long chain molecules
(C) Become infusible on moulding so cannot be reused
(D) Soften on heating and harden on cooling, can be reused

25. Select the correct statements about Ag^+/Ag electrode.

(A) $E_{RP_{Ag^+/Ag}} = E^0_{RP_{Ag^+/Ag}} + \frac{0.059}{1}\log[Ag^+]$

(B) Addition of CN^- in Ag^+ / Ag electrode shows a decrease in the value of $E_{RP_{Ag^+/Ag}}$

(C) Addition of CN^- in Ag^+ / Ag electrode shows a increase in the value of $E_{OP_{Ag/Ag^+}}$

(D) Addition of CN^- in Ag^+ / Ag electrode shows a decrease in $E^0_{OP_{Ag/Ag^+}}$ and an increase in $E^0_{RP_{Ag^+/Ag}}$

26. Mixture of $Ph\text{-}CHO$ & $HCHO$ is treated with $NaOH$ then Cannizzaro reaction involves:
(A) Oxidation of $HCHO$ (B) Hydride transfer from $HCHO$
(C) Oxidation of $Ph - CHO$ (D) Reduction of $Ph - CHO$

Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

List – I		List – II	
(I)	Zero dipole moment	(P)	$BeCl_2$
(II)	Bond angle 180°	(Q)	ClF_3
(III)	Paramagnetic	(R)	CO^+
(IV)	Polar	(S)	PCl_3F_2
		(T)	NO^+

27. Which of the following represent the correct combination?
(A) $I \rightarrow P,\ II \rightarrow S,\ III \rightarrow R,\ IV \rightarrow Q$
(B) $I \rightarrow P,\ II \rightarrow R,\ III \rightarrow S,\ IV \rightarrow Q$
(C) $I \rightarrow S,\ II \rightarrow P,\ III \rightarrow S,\ IV \rightarrow T$
(D) $I \rightarrow S,\ II \rightarrow S,\ III \rightarrow R,\ IV \rightarrow P$

28. Which of the following represent the incorrect combination?
(A) $I \rightarrow P,\ II \rightarrow P,\ III \rightarrow R,\ IV \rightarrow T$
(B) $I \rightarrow R,\ II \rightarrow S,\ III \rightarrow R,\ IV \rightarrow T$
(C) $I \rightarrow S,\ II \rightarrow P,\ III \rightarrow R,\ IV \rightarrow T$
(D) $I \rightarrow P,\ II \rightarrow P,\ III \rightarrow R,\ IV \rightarrow Q$

Space for Rough work

Answer the following by appropriately matching the list based on the information given in the paragraph.

List – I		List – II	
(I)	$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	(P)	$K_p > K_c$ at 300K
(II)	$2NO_2(g) \rightleftharpoons N_2O_4(g)$	(Q)	$K_p < K_c$ at 300K
(III)	$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	(R)	P is increased, reaction moves to right side
(IV)	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	(S)	Inert gas added at constant volume, equilibrium will not be affected
		(T)	Inert gas at constant pressure, equilibrium will shift right

29. Which of the following represent the correct combination?
(A) $I \rightarrow P,\ II \rightarrow Q,\ III \rightarrow Q,\ IV \rightarrow R$
(B) $I \rightarrow Q,\ II \rightarrow R,\ III \rightarrow S,\ IV \rightarrow T$
(C) $I \rightarrow P,\ II \rightarrow Q,\ III \rightarrow S,\ IV \rightarrow R$
(D) $I \rightarrow T,\ II \rightarrow P,\ III \rightarrow Q,\ IV \rightarrow R$

30. Which of the following represent the incorrect combination?
(A) $I \rightarrow P,\ II \rightarrow Q,\ III \rightarrow S,\ IV \rightarrow S$
(B) $I \rightarrow S,\ II \rightarrow Q,\ III \rightarrow S,\ IV \rightarrow S$
(C) $I \rightarrow Q,\ II \rightarrow Q,\ III \rightarrow S,\ IV \rightarrow R$
(D) $I \rightarrow T,\ II \rightarrow Q,\ III \rightarrow S,\ IV \rightarrow R$

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. One mole of nitrogen gas at 0.8 atm takes 38 seconds on diffuse through a pinhole, whereas one mole of a compound of xenon with fluorine at 1.6 atm takes 57 seconds to diffuse through the same hole. Calculate molecular weight of the compound.

32. Calculate $\Delta_f H°$ (magnitude only) for chloride ion (aq.) from the following data:

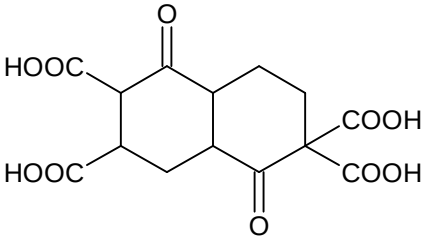
$$\frac{1}{2}H_2(g)+\frac{1}{2}Cl_2(g)\rightarrow HCl(g); \qquad \Delta_f H° = -92.4\,kJ$$

$$HCl(g)+H_2O(\ell)\rightarrow H_3O^+(aq)+Cl^-(aq); \qquad \Delta H° = -74.8\,kJ$$

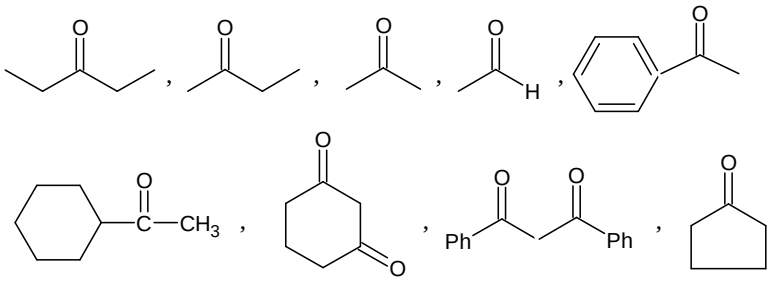
$$\Delta_f H° of\ H^+(aq) = 0.0\,kJ$$

33. K_{sp} of $PbBr_2$ (Molar mass $= 367$) is 3.2×10^{-5}. If the salt is 80% dissociated in solution, calculate the solubility of salt in g per litre.

34. How many moles of CO_2 will released when following compound is heated



35. Examine the structural formulas of compounds given below and identify number of compounds which show positive iodoform test.



36. Calculate the value $X-Y$, for $XeOF_4$. (X=Number of σ bond pair and Y=Number of lone pair on central atom).

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let $A_1, A_2, ..., A_n$ be a regular n-sided polygon inscribed in a circle, three points A_i, A_j, A_k are randomly chosen, where i, j, k are distinct integer between 1 and n both inclusive. Denote by P_n the probability that the triangle $A_iA_jA_k$ is obtuse which of the following statements is/are true:

(A) P_5 equals $\frac{1}{2}$ (B) P_{10} equals $\frac{1}{2}$

(C) P_{10} equals $\frac{2}{3}$ (D) P_{15} equals $\frac{9}{13}$

38. The equation $\frac{1}{(x+1)^5} - 7x + 2\sin x = 0$ has:

(A) One positive root (B) One negative root

(C) None of the root $\in (-1, 0)$ (D) Three real roots

39. Let x, y, z be real numbers with $x \geq y \geq z \geq \frac{\pi}{12}$, such that $x + y + z = \frac{\pi}{2}$ and let $P = \cos x.\sin y.\cos z$, then:

(A) minimum value of P is $\frac{1}{8}$ (B) maximum value of P is $\frac{2+\sqrt{3}}{8}$

(C) minimum value of P is $\frac{1}{4}$ (D) maximum value of P is $\frac{2+\sqrt{3}}{4}$

Space for Rough work

40. The equation of two lines making an angle 45° with a given line $\bar{a}z + a\bar{z} + b = 0$ (where 'a' is a complex number and b is real) and passing through a given point c (c is a complex number), is/are:

(A) $\frac{z+c}{a} + i\left(\frac{\bar{z}-\bar{c}}{a}\right) = 0$ (B) $\frac{z-c}{a} + i\frac{\bar{z}-\bar{c}}{\bar{a}} = 0$

(C) $\frac{z-c}{a} - i\left(\frac{\bar{z}-\bar{c}}{\bar{a}}\right) = 0$ (D) $\frac{z+c}{a} - i\frac{\bar{z}-\bar{c}}{\bar{a}} = 0$

41. If the equation $|x^2-9| + |x^2-5| = |\lambda - 5|$ has infinite solution, then λ is:

(A) 9 (B) 1
(C) 4 (D) 2

42. Five circles $S_i = 0, i = 1,2,3,4,5$ have centres respectively at $C_1(0,0), C_2(2,2), C_3(6,10), C_4(1,-2), C_5(4,-3)$ and the line $y = mx + c$ is common tangent to all five circles such that C_1, C_2, C_3 lie on one side and C_4, C_5 lie on other side of the tangent it is given that the sum of radii of $S_1 = 0$ and $S_2 = 0$ is equal to radius of $S_4 = 0$ and also sum of radii of $S_2 = 0$ and $S_3 = 0$, is equal to the radius of $S_5 = 0$, then:

(A) $m = c$ (B) $m + c = 0$
(C) $m^2 + c^2 = 2$ (D) $|m| = |c|$

43. If $\begin{vmatrix} 2bc-a^2 & c^2 & b^2 \\ c^2 & 2ca-b^2 & a^2 \\ b^2 & a^2 & 2ab-c^2 \end{vmatrix} = \left(a^3+b^3+c^3+kabc\right)^2$, then k is equal to

(A) 2 (B) -2
(C) 3 (D) -3

Space for Rough work

44. If $\sum_{i=0}^{\infty}\sum_{j=0}^{\infty}\frac{1}{a^i a^j}=\frac{\lambda a^2}{(a-1)^2(a+1)}$ where $a>1$, then possible values of λ may be:

(A) 1 (B) 2

(C) 3 (D) 4

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five/Six** entries (P), (Q), (R), (S), (T) or (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

If e_1 and e_2 are the roots of the equation $x^2-ax+2=0$, then there are same expressions given in the list – I whose values are given in list – II below:

List – I		List – II	
(I)	If e_1 and e_2 are the eccentricities of the ellipse and hyperbola respectively, then the possible values of 'a' is/are	(P)	5
(II)	If both e_1 and e_2 are the eccentricities of hyperbola then the possible value of a is/are	(Q)	$\frac{5}{2}$
(III)	If e_1 and e_2 are eccentricities of hyperbola and its conjugate hyperbola, then the possible values of a is/are	(R)	6
(IV)	If e_1 is the eccentricity of the hyperbola for which there exist infinite points from which perpendicular tangents can be drawn and e_2 is the eccentricity of the hyperbola from which no such points exists then the possible of 'a' is/are	(S)	$2\sqrt{2}$

45. Which of the following is the only correct combination?

(A) $\text{IV}\rightarrow\text{QS}$ (B) $\text{II}\rightarrow\text{PR}$

(C) $\text{I}\rightarrow\text{PR}$ (D) $\text{III}\rightarrow\text{Q}$

Space for Rough work

46. Which of the following is the only incorrect combination?

(A) $\text{II} \rightarrow \text{QS}$ (B) $\text{II} \rightarrow \text{PR}$

(C) $\text{I} \rightarrow \text{PR}$ (D) $\text{IV} \rightarrow \text{PR}$

Answer the following by appropriately matching the list based on the information given in the paragraph.

The equation of the ellipse with area $\frac{3\text{P}}{2}$ passing through the point of intersection of $\text{x}^2 + 4y^2 = 4$ and $x - 1 = 0$, having x-axis as its major axis and eccentricity equal to $\frac{\sqrt{3}}{2}$, is $ax^2 + 2hxy + by^2 + 2gx + 2fy - 2 = 0$. There are same expressions given in the list – I whose values are given in list – II below

List – I		List – II	
(I)	a	(P)	0
(II)	b	(Q)	1
(III)	$2h + f$	(R)	-1
(IV)	g	(S)	4
		(T)	6
		(U)	$\sqrt{6}$

47. Which of the following is the only correct combination?

(A) $\text{II} \rightarrow \text{P}$ (B) $\text{II} \rightarrow \text{S}$

(C) $\text{I} \rightarrow \text{T}$ (D) $\text{I} \rightarrow \text{R}$

48. Which of the following is the only incorrect combination?

(A) $\text{IV} \rightarrow \text{U}$ (B) $\text{III} \rightarrow \text{P}$

(C) $\text{II} \rightarrow \text{S}$ (D) $\text{I} \rightarrow \text{Q}$

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Mr. A lists all the positive divisors of the number $N(2010)^2$ and selects two divisors from the list then the probability that exactly one of the selected divisors is perfect squares is:

50. If $\frac{(x-3)^{\frac{-|x|}{x}}\sqrt{(x-4)^2}\,(17-x)}{\sqrt{-x}\left(-x^2+x-1\right)\left(|x|-32\right)} < 0$ then no. of integers x satisfying the inequality is:

51. Let f be real function defined on R (the set of real numbers) such that $f'(x) = 100\,(x-1)(x-2)^2(x-3)^3.....(x-100)^{100}$, for all $x \in R$. If g is a function defined on R such that $\int_a^x e^{f(t)}dt = \int_0^x g(x-t)dt + 2x + 3$, If some of the all the values of x for which g(x) has a local extremum be λ then find $\frac{\lambda}{3}$

52. Let z_1, z_2 and z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = |z_1+z_2+z_3| = 2$. If $|z_1 - z_2| = |z_1 - z_3|$ *and* $z_2 \neq z_3$. Then find the value of $|z_1+z_2||z_1+z_3|$.

53. If $\lim_{x\to 1}\frac{100}{1-x^{100}} - \frac{50}{1-x^{50}} = 5A$; find the value of A.

54. Let $H_n = 1+\frac{1}{2}+\frac{1}{3}+...+\frac{1}{n}$ and $T_n = \frac{1}{(n+1)H_nH_{n+1}}$, then the value of $(T_1+T_2+T_3+...+\infty)$ is:

Space for Rough work